

EquiPhase

Track-1 Implementation Brief + Staged Antigravity Prompts

Deep-Equilibrium Benchmarking for Condition-Dependent Phase Separation and Inverse Design

Hand this entire document to Antigravity as the project context. The agent reads Sections 0-9, then executes the prompts in the Appendix in order, pausing at each review gate.

READ FIRST: this project succeeds only if every number is a live measurement. Recycled, copied, or assumed values are failures, not shortcuts.

0. Purpose and how to use this document

EquiPhase tests one question: does a deep-equilibrium (implicit fixed-point) model beat established baselines at predicting and designing condition-dependent protein phase separation, specifically when extrapolating to unseen conditions? The result is publishable in a computational venue if the hypothesis holds, and is reported honestly (including as a null result) if it does not. Antigravity executes; the human author (with an external reviewer) audits at each review gate.

1. The single hypothesis (H1)

H1: a DEQ formulation predicts the condition-dependent phase boundary / saturation concentration better than feed-forward, equilibrium-distribution (DiG-style), and active-learning baselines, with its advantage concentrated in the out-of-condition (extrapolation) regime. The benchmark is designed so H1 can be cleanly falsified.

2. Standing integrity rules (apply to EVERY prompt)

These are non-negotiable and override any instinct to make results look better. They exist because a prior pipeline in this line of work emitted a recycled convergence log.

- R1 — Pre-registration is immutable after freeze. Changing any frozen target, split, or metric requires explicit human approval, recorded as a dated deviation in DEVIATIONS.md.
- R2 — Report only measured values from a live run, each tagged with a unique run_id and a config hash. Never copy a number between runs or experiments.
- R3 — Anti-recycling. Every convergence/residual trace is regenerated per run and saved. If any new run's residual trace is identical to a different run or system, HALT and report it as a fabrication signal.
- R4 — Single source of truth. Any derived quantity (gap, CI, enrichment, K-equivalent) is computed once by a named function in metrics.py; no hand-entered values anywhere in code, logs, or write-ups.
- R5 — Failures and nulls are valid outputs. Do not smooth traces, hide diverged runs, or retune against the locked-test. A null result is reported as a null result.
- R6 — Leakage control. The locked-test set is opened exactly once, after freeze, by a single script. No test-set information touches training or model selection.
- R7 — Ask, don't invent. If data, licences, or specs are ambiguous, stop and ask the human rather than guessing or fabricating.

3. Task and data

- Primary task: (a) binary LLPS / no-LLPS classification, and (b) saturation-concentration (or phase-boundary) regression, conditioned on environmental variables (concentration, salt, temperature, pH, crowding).
- Candidate public sources (verify licence + current release before use): LLPSDB v2 (condition-annotated in-vitro LLPS), PhaSePred / DrLLPS / CD-CODE / PhaSepDB. Keep every record's source ID.
- Splits: by Bemis-Murcko-equivalent protein-family clustering AND by condition regime (e.g., train low-salt, test high-salt) AND a temporal embargo where assay dates exist. Three split files: train / val / locked-test.

4. Baselines and ablations

Model	Represents	Role
ESM-2 embedding + MLP	Feed-forward sequence baseline	Does equilibrium structure help at all?
Condition-aware ML (LLPSense-style)	PLM + environment variables	Current SOTA for condition dependence
DiG-style equilibrium-distribution	Learns equilibrium distribution; bias for design	Closest 'equilibrium + inverse design' rival
Active-ML navigator	Iterative sample selection	Beats smart sampling, not just static models?
catGRANULE / FuzDrop	Heuristic predictors	Sanity floor
DEQ (candidate)	Implicit fixed-point, O(1) memory	Method under test
Ablation A: unrolled finite depth	Removes implicit differentiation	Value of the implicit-function gradient
Ablation B: no condition coupling	Removes self-consistent condition dependence	Value of the equilibrium inductive bias

5. Metrics and the decisive metric

- In-distribution: AUROC / AUPRC (classification); Spearman rho, RMSE (regression).

Decisive: extrapolation gap = (in-distribution score) minus (out-of-condition score), per model. H1 predicts the DEQ has the smallest gap.

- Efficiency: peak VRAM vs effective depth; wall-clock on one RTX 4090. Report DEQ's O(1)-memory profile explicitly.

6. Pre-registered win condition (freeze before any training)

DEQ is declared advantageous only if ALL hold: (i) beats the best baseline on the extrapolation metric by Spearman ≥ 0.05 with non-overlapping 95% bootstrap CIs across ≥ 5 seeds; (ii) matches or beats the best baseline in-distribution; (iii) the memory advantage is quantified. Win

only on extrapolation + memory (tie in-distribution) is a legitimate, reportable story. Loss across the board => null-result note. No re-tuning.

7. Inverse-design sub-benchmark

With each model frozen, generate sequence mutations predicted to lower the saturation concentration: DEQ by gradient design, DiG by biased sampling, active-ML by propose-and-rank. Score enrichment (EF@k / BEDROC) against held-out, experimentally known LLPS-promoting mutations (e.g., FUS / hnRNPA1 / tau disease mutants). Retrospective only — no wet lab — so it stays honest while testing 'can the DEQ design, not just predict'.

8. Compute and repo conventions

- Single RTX 4090. Python venv; pin versions in requirements.txt. DEQ via a maintained implicit-layer library (e.g., torchdeq) — verify availability.
- Repo: equiphase/ with data/, splits/, models/, baselines/, eval/, configs/, runs/<run_id>/, reports/. Every run writes config hash, seed, git SHA, full logs.
- One shared evaluation harness so all models are scored identically. No model gets a bespoke metric.

9. Deliverables and review gates

After each Appendix prompt, Antigravity stops at a REVIEW GATE and returns artefacts for the author + reviewer to audit before the next stage. Deliverables: IMPLEMENTATION_PLAN.md, PRE_REGISTRATION.json (hashed), the three split files + manifest, baseline results, DEQ + ablation results, extrapolation + inverse-design results, and a final honest report (PASS / PARTIAL / NULL).

Appendix — Staged Antigravity Prompts

Paste these one at a time. Do not advance past a REVIEW GATE without author sign-off. The standing rules (Section 2) apply to all.

Standing-rules reminder block (prepend to any prompt if the agent drifts):

REMINDER: Report only live-measured values tagged with run_id + config hash. Never copy a number between runs. Regenerate every residual trace; identical traces => HALT and report. Compute derived quantities once in metrics.py. Failures and nulls are valid; do not smooth, hide, or retune to win. Locked-test opens once. If anything is ambiguous, ask the human; do not invent.

— End of brief —